

Pointwise Convergence for Two Commuting Transformations

Proof manuscript

Abstract

We present a proof argument for pointwise convergence of bilinear averages associated with two commuting invertible measure-preserving transformations. Finite-matrix and heat-flow lemmas yield a smooth kernel estimate through higher Gaussian derivatives and an entrywise switching argument. A planar maximal estimate, spatial transference, closure in the kernel's L^1 norm, and exact phase reconstruction then give convergence for bounded functions. Truncation extends the conclusion to $L^2 \times L^2$. This is a proof manuscript for review; independent verification of the analytic argument and its Lean formalization remain incomplete.

1 Main statement and planar estimate

Let $(X, \mathcal{B}, \mu)$ be a probability space. Let T, S be invertible, measurable, measure-preserving transformations with measurable inverses and $TS = ST$. For measurable complex-valued functions f, g and $N \geq 1$, put

$$A_N(f, g)(x) = \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) g(S^n x).$$

Conjecture. For every such system and every $f, g \in L^\infty(\mu)$, the sequence $A_N(f, g)(x)$ converges for μ -almost every x .

Set

$$\varphi(t) = (2\pi)^{-1/2} e^{-t^2/2}, \quad E(t) = (3 - t^2)\varphi(t), \quad O(t) = (3t - t^3)\varphi(t),$$

and write $D_r K(t) = r^{-1} K(t/r)$ for $r > 0$. For $K \in L^1(\mathbb{R})$ and bounded measurable $F, G : \mathbb{R}^2 \rightarrow \mathbb{C}$, define

$$\mathcal{A}_N^K(F, G)(u, v) = \int_{\mathbb{R}} K(t) F(u + Nt, v) G(u, v + Nt) dt.$$

An m -chain is a tuple $\mathbf{n} = (n_0, \dots, n_m)$ of positive integers with $n_0 < \dots < n_m$. For a sequence $a = (a_N)$ set

$$V_{\mathbf{n}}(a) = \sum_{j=1}^m |a_{n_j} - a_{n_{j-1}}|, \quad M_{\mathcal{F}}(a) = \max_{\mathbf{n} \in \mathcal{F}} V_{\mathbf{n}}(a),$$

where $\mathcal{F}$ is a finite nonempty family of m -chains. Let $Q_L = [-L, L]^2$, with $|Q_L| = 4L^2$.

Proposition 1 (Planar finite maximal estimate). *For each $K \in \{D_r E, D_r O : r > 0\}$ and $C_f, C_g \geq 0$, there are finite numbers $C_m \geq 0$, $m \geq 1$, such that $C_m/m \rightarrow 0$ and*

$$\int_{\mathbb{R}^2} M_{\mathcal{F}}((\mathcal{A}_N^K(F, G)(p))_{N \geq 1}) dp \leq |Q_L| C_m \quad (1)$$

for every $L > 0$, every bounded measurable F, G vanishing outside Q_L with $|F| \leq C_f$, $|G| \leq C_g$, and every finite nonempty family $\mathcal{F}$ of m -chains. The sequence (C_m) is independent of $L, F, G, \mathcal{F}$.

Proof. Proposition 3 establishes the smooth kernel estimate from the finite-matrix and heat-flow lemmas stated in Section 2. Lemma 4 converts that estimate into (1), with $C_m = C_\kappa C_f C_g \sqrt{m}$ for $K = D_r \kappa$, $\kappa = E, O$. In particular $C_m/m \rightarrow 0$. $\square$

The maximum in (1) is inside the integral. The proof therefore requires an estimate uniform over output-dependent choices of chains, as supplied by the switching argument in Section 2.

Theorem 2. *For every probability space and pair T, S as above, and every $f, g \in L^2(\mu)$, the sequence $A_N(f, g)$ converges almost everywhere. In particular, the conclusion holds for $f, g \in L^\infty(\mu)$ and for $f, g \in L^3(\mu)$.*

We prove this theorem below. Throughout, representatives may be chosen measurable; a bounded equivalence class may be represented by a function bounded everywhere. Changing representatives does not affect the discrete averages outside the countable union of the inverse images of their exceptional sets under T^n and S^n .

2 The analytic estimate

We use the finite-matrix and heat-flow lemmas stated below. Their proofs are external inputs and are not reproduced in this manuscript. We give the higher-derivative and switching argument that derives the planar estimate from these lemmas. The status of the mathematical review and the status of Lean formalization are separate from the logical dependencies of this written argument.

Write $g_\sigma(t) = (2\pi\sigma)^{-1/2} e^{-t^2/(2\sigma)}$. For finite arrays A_v , with indices v taken modulo 3, use a mesh $h > 0$, nodes $u_i = ih$, and weights

$$w_v(i) = h g_{\sigma_v}(u_i - p_v), \quad W_v(i, j) = A_v(i, j) \sqrt{w_v(i) w_{v+1}(j)}, \quad n_v = \left(\sum_{i,j} h^2 |A_v(i, j)|^3 \right)^{1/3}.$$

Put $R_v = [W_v \ W_{v-1}^*]$, $T_v = R_v^* R_v$, $S_v = R_v R_v^*$, and

$$\Phi(P) = \text{tr}(P^{3/2}), \quad b_P(K) = \frac{3}{2} \sum_{i,j} \frac{|K_{ij}|^2}{\sqrt{p_i} + \sqrt{p_j}}.$$

Here the sum is taken in an eigenbasis of the positive support of P , and K is Hermitian and supported there. We retain the following matrix inputs: joint convexity of $(P, K) \mapsto b_P(K)$ under probability averages with a fixed support and integrable operator norms; and

$$|\text{tr}(D_0 W_0 D_1 W_1 D_2 W_2)| \leq \frac{5}{3} \max_v \|D_v\|_{\text{op}} \sum_v b_{T_v}(R_v^* D_v R_v) \quad (2)$$

for Hermitian D_v . We also retain the rank-change differentiation formula

$$\Phi(R_+^* R_+) - \Phi(R_-^* R_-) = 3 \int_0^1 \operatorname{Re} \langle R_z \sqrt{R_z^* R_z}, R_+ - R_- \rangle_{\text{HS}} dz$$

with $R_z = R_- + z(R_+ - R_-)$, and the entry bound

$$|(R \sqrt{R^* R})_{ij}| \leq \|\operatorname{row}_i R\|_2 \|\operatorname{col}_j R\|_2.$$

Let $C_{v,w}$ denote the block of R_v belonging to its neighbor w . Set

$$N_v = \operatorname{diag}_i \frac{u_i - p_v}{\sigma_v}, \quad U_v = R_v^* N_v R_v, \quad V_v = b_{T_v}(U_v),$$

$$P_{v,w} = C_{v,w} N_w C_{v,w}^*, \quad Y_{v,w} = b_{S_v}(P_{v,w}).$$

The polarization of b_{S_v} applied to the two column insertions is denoted Z_v ; in particular $2Z_v \leq \sum_{w \neq v} Y_{v,w}$. Let $\Pi_d = \{p_0 + p_1 + p_2 = d\}$, with any two centers as free Lebesgue coordinates, and $J_v(d, s) = \int_{\Pi_d} \Phi(T_v) d\pi_d$, where $\sigma_i = \alpha_i s$. The heat-flow inputs are

$$\begin{aligned} 2\partial_s J_v &= \left(\sum_i \alpha_i \right) \partial_d^2 J_v - \int_{\Pi_d} \left(\alpha_v V_v + \sum_{w \neq v} \alpha_w Y_{v,w} \right), \\ \partial_d^2 J_v &= \int_{\Pi_d} Z_v. \end{aligned} \tag{3}$$

We retain the initial bound $J_v \leq C \sum_i n_i^3$ whenever all $\sigma_i \geq h^2$. For a single retained edge, its energy $J_{v,w}^e$ is independent of d and obeys

$$-2\partial_s J_{v,w}^e = \iint (\alpha_v V_{v,w}^e + \alpha_w Y_{v,w}^e) dp_v dp_w, \quad Y_{v,w} \leq Y_{v,w}^e.$$

Its initial energy is bounded by the cube of that edge's discrete L^3 norm. Gaussian domination of the needed center derivatives on compact positive scale intervals is part of these inputs.

Proposition 3 (Smooth kernel estimate). *For bounded measurable compactly supported F, G and $\kappa = E, O$,*

$$\|\mathcal{C}_m(B^\kappa(F, G))\|_{3/2} \leq C_\kappa \sqrt{m} \|F\|_3 \|G\|_3, \tag{4}$$

where

$$B_L^\kappa(F, G)(x, y) = \int_{\mathbb{R}} F(x+t, y) G(x, y+t) D_L \kappa(t) dt,$$

and $\mathcal{C}_m$ is the pointwise supremum of sums of increments over at most m ordered, nonoverlapping positive scale intervals. The bound is uniform over finite menus and output-dependent choices.

Proof. Put $s = L^2$ and $k_{r,L} = L^r g_{s/3}^{(r)}$. The ratio $k_{r,L}/g_s$ is bounded uniformly in L , for each fixed r . Narrow the row variance at vertex v to $s/6$, move its center by d , and keep its column weights fixed. Denote the new Gram and score insertion by T_v^d, U_v^d . With $d\nu_s(d) = g_{5s/6}(d) dd$, Gaussian convolution gives

$$T_v^{\text{base}} = \int T_v^d d\nu_s, \quad R_v^* \operatorname{diag}(k_{r,L}/g_s) R_v = \int c_{r,s}(d) U_v^d d\nu_s,$$

$$c_{r,s}(d) = -\frac{L^r g_{s/6}^{(r-1)}(d)}{g_{5s/6}(d)}, \quad |c_{r,s}(d)| \leq C_r \sqrt{s}.$$

All row weights are positive, so the Gram support is independent of d . In finite dimensions the relevant operator norms are integrable against ν_s . Joint convexity and quadratic homogeneity therefore give

$$b_{T_v^{\text{base}}}(R_v^* \text{diag}(k_{r,L}/g_s) R_v) \leq C_r s \int V_v^d d\nu_s. \quad (5)$$

The signed coefficient remains in the insertion; it is not used as a probability density.

For this narrowed vertex define $\tilde{J}_v = \int J_v(d, s) d\nu_s(d)$ and use a tilde for the corresponding averaged costs. The variance rates sum to $13/6$. Differentiating the averaging density adds heat rate $5/6$. Thus (3) and integration by parts give

$$2\partial_s \tilde{J}_v = 3\tilde{Z}_v - \frac{1}{6}\tilde{V}_v - \sum_{w \neq v} \tilde{Y}_{v,w},$$

whence

$$\frac{1}{6}\tilde{V}_v \leq -2\partial_s \tilde{J}_v + \frac{1}{2} \sum_{w \neq v} \tilde{Y}_{v,w}. \quad (6)$$

The last terms are controlled by single-edge dissipation: their plane integrals do not depend on d , and each column rate is 1.

Now change only edge 0, using at most m nonoverlapping scale intervals and coefficients of modulus at most 1 at each (i, j) :

$$B_0^s(i, j) = A_0(i, j) \sum_{\ell} \beta_{\ell}(i, j) \mathbf{1}_{\{a_{\ell}(i, j)^2 < s < b_{\ell}(i, j)^2\}}.$$

Then $|B_0^s| \leq |A_0|$ and $\sum_{\tau} |\Delta B_0^{\tau}(i, j)| \leq 2m|A_0(i, j)|$. At a switch, use the original arrays to define

$$\begin{aligned} X^2 &= \sum_l |A_0(i, l)|^2 w_1(l), & Y^2 &= \sum_l |A_0(l, j)|^2 w_0(l), \\ H^2 &= \sum_l |A_2(l, i)|^2 w_2(l), & Q^2 &= \sum_l |A_1(j, l)|^2 w_2(l). \end{aligned}$$

The rank-change formula and the row-column bound yield

$$|\Delta e_0| + |\Delta e_1| \leq C \sum_{i,j} w_0(i) w_1(j) |\Delta B_0(i, j)| (2XY + HY + QX),$$

where $e_v = \Phi(T_v)$. Star 2 does not change. For fixed i, j , use the probability measure

$$h^{-2} w_0(i) w_1(j) d\pi_d d\nu_s(d).$$

Gaussian convolution gives

$$\begin{aligned} \mathbb{E} w_0(l) &= h g_{2\alpha_0 s}(u_l - u_i), & \mathbb{E} w_1(l) &= h g_{2\alpha_1 s}(u_l - u_j), \\ \mathbb{E} w_2(l) &= h g_{3s}(u_l + u_i + u_j). \end{aligned}$$

Gaussian averages are bounded by the discrete one-dimensional maximal operator, uniformly when the variances exceed the mesh scale. Hence $\mathbb{E} X^2, \mathbb{E} Y^2, \mathbb{E} H^2, \mathbb{E} Q^2$ have scale-independent majorants whose square roots have discrete L^3 norms at most Cn_0, Cn_0, Cn_2, Cn_1 , respectively. This follows by the strong $L^{3/2}$ maximal inequality on each slice; the index $-i-j$ is a bijection along the relevant slice.

Cauchy-Schwarz for the center probability measure, the total entrywise switch bound, and discrete Hölder now give

$$\sum_{\tau} \sum_v |\Delta \tilde{J}_v| \leq C m \{n_0^3 + n_0^2(n_1 + n_2)\}.$$

For the varying single edge the same argument costs Cmn_0^3 . Fixed edges have no jumps and are not initialized anew at each switch. Integrating (6) between switches and telescoping the energies therefore yields

$$\int \tilde{V}_v(s) ds \leq C \left[\sum_{j=0}^2 n_j^3 + m\{n_0^3 + n_0^2(n_1 + n_2)\} \right]. \quad (7)$$

All scales lie in a fixed compact interval during this step, and the mesh is chosen with $h^2 \leq s_-/6$.

For $r \geq 3$, insert the three derivative windows with orders $(1, 1, r-2)$ into (2). Apply (5) and integrate over the center plane. Their convolution is

$$k_{1,L} * k_{1,L} * k_{r-2,L} = L^r g_s^{(r)} = D_L \varphi^{(r)}.$$

Integration with $ds/s = 2dL/L$ and (7) bounds the trilinear form for output-dependent sums of kernels

$$I_{a,b}^{(r)} = \int_a^b D_L \varphi^{(r)} \frac{dL}{L}.$$

Normalize the three input norms to $(m^{-1/2}, 1, 1)$. The bracket in (7) becomes $m^{-3/2} + 2 + m(m^{-3/2} + 2m^{-1}) \leq 6$. Restoring the original norms by trilinearity gives the bound $C_r \sqrt{m} \prod_v n_v$.

Fix a finite scale menu and a rectangular step choice before refining the mesh. Riemann sums then yield the corresponding continuous trilinear estimate. To pass to measurable choices, keep the valid interval lists as finitely many labels and approximate their measurable partition by rectangular partitions in the finite measure having density the sum of the absolute trilinear integrands. Every label preserves the switch bound. For each fixed menu all kernels are integrable; Hölder also gives continuity in the three L^3 inputs, so density extends the estimate to those inputs. Finite menus of rational endpoints and a countable dense set of phases are then increased monotonically.

Finally, direct differentiation gives

$$\partial_{\log L} D_L E = -D_L \varphi^{(4)}, \quad \partial_{\log L} D_L O = D_L(\varphi^{(5)} + 3\varphi^{(3)}).$$

Use the estimates of orders 3, 4, 5, linearize each absolute increment by a complex phase, and use $L^{3/2}$ – L^3 duality. The substitution $x = -a, y = -b, t = a + b + c$ identifies the triangular form with B_L^κ and preserves the input norms. This proves (4) from the stated finite-matrix and heat-flow lemmas. $\square$

Lemma 4. *The smooth estimate (4) implies the planar estimate (1) with $C_m = C_\kappa C_f C_g \sqrt{m}$ for $K = D_r \kappa$, $\kappa = E, O$.*

Proof. For every N , changing variables gives $\mathcal{A}_N^{D_r \kappa}(F, G) = B_{rN}^\kappa(F, G)$. Every finite chain maximum is bounded by $\mathcal{C}_m(B^\kappa)$. If F, G vanish outside Q_L , these averages vanish outside Q_L : $F(u+t, v)$ forces $|v| \leq L$, and $G(u, v+t)$ forces $|u| \leq L$. Hölder therefore gives

$$\begin{aligned} \int_{\mathbb{R}^2} M_{\mathcal{F}}(\mathcal{A}^{D_r \kappa}) &\leq |Q_L|^{1/3} \|\mathcal{C}_m(B^\kappa)\|_{3/2} \\ &\leq |Q_L|^{1/3} C_\kappa \sqrt{m} \|F\|_3 \|G\|_3 \leq |Q_L| C_\kappa C_f C_g \sqrt{m}. \end{aligned}$$

The constants do not depend on the finite family or on L . Since $\sqrt{m}/m \rightarrow 0$, the constants have the required sublinear growth. $\square$

3 A convergence criterion and spatial transference

Lemma 5. *Let (Y, ν) be a probability space and $a_N : Y \rightarrow \mathbb{C}$ measurable. Suppose that $\int_Y M_{\mathcal{F}}(a) d\nu \leq C_m$ for every finite family of m -chains, where $C_m < \infty$ and $C_m/m \rightarrow 0$. Then a_N converges almost everywhere.*

Proof. The set of m -chains is countable. Enumerating it and applying monotone convergence to finite maxima gives

$$J_m(y) := \sup_{\mathbf{n}} V_{\mathbf{n}}(a(y)), \quad \int_Y J_m d\nu \leq C_m.$$

If a sequence is not Cauchy, there is $\eta > 0$ such that arbitrarily far out two terms have distance at least 2η . From any chosen term, at least one of these two later terms has distance at least η . Induction therefore constructs chains of every length with all successive increments at least η . Thus $J_m \geq m\eta$ for every m . For each fixed rational $\eta > 0$, Markov's inequality gives

$$\nu \left(\bigcap_{m \geq 1} \{J_m \geq m\eta\} \right) \leq \inf_{m \geq 1} \frac{C_m}{m\eta} = 0.$$

Taking a countable union in η proves the Cauchy property almost everywhere. Completeness of $\mathbb{C}$ gives convergence. $\square$

Let U_t, V_t be commuting, jointly measurable, measure-preserving $\mathbb{R}$ -actions on a probability space (Y, ν) . For bounded measurable $f, g : Y \rightarrow \mathbb{C}$, put

$$a_N^K(y) = \int_{\mathbb{R}} K(t) f(U_{Nt}y) g(V_{Nt}y) dt. \quad (8)$$

The integral is absolutely convergent and defines a measurable function.

Proposition 6. *Fix $K \in L^1(\mathbb{R})$ and bounds $|f| \leq C_f$, $|g| \leq C_g$. If (1) holds for this kernel, these bounds, and a fixed m with constant C_m , then*

$$\int_Y M_{\mathcal{F}}((a_N^K(y))_{N \geq 1}) d\nu(y) \leq C_m$$

for every finite nonempty family $\mathcal{F}$ of m -chains.

Proof. Let B be the largest endpoint in $\mathcal{F}$. For $y \in Y$ define

$$F_y(u, v) = f(U_u V_v y), \quad G_y(u, v) = g(U_u V_v y).$$

Commutativity gives the exact identity

$$\mathcal{A}_N^K(F_y, G_y)(p) = a_N^K(U_{p_1} V_{p_2} y).$$

Fix $R, L > 0$ and replace F_y, G_y by their restrictions to Q_{L+BR} , extended by zero. Denote the resulting functions by F'_y, G'_y . For $p \in Q_L$, $N \leq B$, and $|t| \leq R$, both shifted points in the integrand remain in Q_{L+BR} . Consequently

$$|\mathcal{A}_N^K(F_y, G_y)(p) - \mathcal{A}_N^K(F'_y, G'_y)(p)| \leq \varepsilon_R, \quad \varepsilon_R = 2C_f C_g \int_{|t| > R} |K(t)| dt.$$

Each chain sum changes by at most $2m\varepsilon_R$. The same bound holds for the finite maximum, independently of the size of $\mathcal{F}$. Applying (1) to F'_y, G'_y yields

$$\int_{Q_L} M_{\mathcal{F}}(a^K(U_{p_1} V_{p_2} y)) dp \leq |Q_{L+BR}| C_m + 2m\varepsilon_R |Q_L|.$$

Integrate in y . Tonelli's theorem and measure preservation identify the left side with $|Q_L| \int_Y M_{\mathcal{F}}(a^K) d\nu$. Division by $|Q_L| > 0$ gives

$$\int_Y M_{\mathcal{F}}(a^K) d\nu \leq \left(1 + \frac{BR}{L}\right)^2 C_m + 2m\varepsilon_R.$$

First let $L \rightarrow \infty$ with R fixed, then let $R \rightarrow \infty$. Integrability of K proves the claim. $\square$

Corollary 7. *The averages (8) converge almost everywhere for each $K = D_r E, D_r O, r > 0$, and bounded measurable f, g .*

Proof. Proposition 1 supplies the finite maximal estimate. Apply Proposition 6 for every m , followed by Lemma 5. $\square$

4 Passage from Gaussian kernels to the interval kernel

Lemma 8. *Let $H_N(y, t)$ be measurable complex-valued functions on $Y \times \mathbb{R}$ with $|H_N(y, t)| \leq M$ for every N, y, t . The real kernels $K \in L^1(\mathbb{R})$ for which $\int_{\mathbb{R}} K(t) H_N(y, t) dt$ converges for almost every y form a closed linear subspace $\mathcal{C}$ of $L^1(\mathbb{R}; \mathbb{R})$.*

Proof. Linearity follows by intersecting finitely many sets of full measure. Suppose $K_j \in \mathcal{C}$ and $\|K_j - K\|_1 \rightarrow 0$. Intersect the countably many sets of convergence for K_j . At any remaining point,

$$\sup_N \left| \int_{\mathbb{R}} (K - K_j)(t) H_N(y, t) dt \right| \leq M \|K - K_j\|_1.$$

The Cauchy criterion, first in N and then in j , proves convergence for K . This proves closedness. $\square$

Lemma 9. *If a closed linear subspace $\mathcal{C} \subset L^1(\mathbb{R}; \mathbb{R})$ contains $D_r E$ and $D_r O$ for every $r > 0$, it contains $t^d e^{-at^2}$ for every integer $d \geq 0$ and every $a > 0$.*

Proof. For $k = 0, 1$ put

$$E_{k,a}(t) = (3 - 2at^2)t^k e^{-at^2}, \quad G_{k,a}(t) = t^k e^{-at^2}.$$

Each $E_{k,a}$ is a constant multiple of a positive dilation of E or O , hence lies in $\mathcal{C}$. The map $s \mapsto E_{k,a}(s \cdot)$ is continuous in L^1 on every compact subinterval of $(0, \infty)$: on $[\delta, 1]$ a common integrable bound is a constant times $(|t|^k + |t|^{k+2})e^{-a\delta^2 t^2}$, so dominated convergence applies. For $0 < \delta < 1$, integration of the derivative of $s^3 t^k e^{-as^2 t^2}$ gives

$$\int_{\delta}^1 s^{2-k} E_{k,a}(st) ds = G_{k,a}(t) - \delta^3 t^k e^{-a\delta^2 t^2}. \quad (9)$$

The left side is an L^1 limit of Riemann sums of elements of $\mathcal{C}$. Here $E_{k,a}(st) = s^k E_{k,as^2}(t)$, so every summand belongs to $\mathcal{C}$. The remainder satisfies the exact identity

$$\left\| \delta^3 t^k e^{-a\delta^2 t^2} \right\|_1 = \delta^{2-k} \|G_{k,a}\|_1 \longrightarrow 0.$$

It follows that $G_{0,a}, G_{1,a} \in \mathcal{C}$ for every $a > 0$.

For $h > 0$ and an integer $j \geq 0$, consider

$$t^k e^{-at^2} \left(\frac{1 - e^{-ht^2}}{h} \right)^j = h^{-j} \sum_{\ell=0}^j (-1)^\ell \binom{j}{\ell} t^k e^{-(a+\ell h)t^2}.$$

The right side lies in $\mathcal{C}$. Since $0 \leq (1 - e^{-ht^2})/h \leq t^2$ and this quotient tends to t^2 , dominated convergence, with bound $|t|^{k+2j} e^{-at^2}$, gives the L^1 limit $t^{k+2j} e^{-at^2}$. Every nonnegative degree is of the form $k + 2j$ with $k = 0$ or 1 . $\square$

Lemma 10. *The function $\chi = \mathbf{1}_{(0,1)}$ is an $L^1(\mathbb{R})$ limit of functions $P(t)\varphi(t)$ with P a real polynomial.*

Proof. Let $d\gamma(t) = \varphi(t) dt$. We first show that real polynomials are dense in $L^2(\gamma; \mathbb{R})$. If h is orthogonal to all polynomials, the function

$$F(z) = \int_{\mathbb{R}} h(t) e^{zt} d\gamma(t), \quad z \in \mathbb{C},$$

is entire. Indeed, on every bounded set of z its integrand and all its derivatives are dominated by integrable functions, by Cauchy–Schwarz and the Gaussian exponential moments. All derivatives of F at zero vanish by orthogonality, so F is identically zero. In particular the Fourier transform of the finite signed measure $h d\gamma$ vanishes. Uniqueness of the Fourier transform for finite measures gives $h = 0$ almost everywhere. The orthogonal-complement criterion proves density.

Now $\chi/\varphi \in L^2(\gamma)$, since

$$\int_{\mathbb{R}} \left| \frac{\chi}{\varphi} \right|^2 d\gamma = \int_0^1 \frac{dt}{\varphi(t)} < \infty.$$

Choose polynomials $P_j \rightarrow \chi/\varphi$ in $L^2(\gamma)$. As $\gamma(\mathbb{R}) = 1$, Cauchy–Schwarz gives

$$\|P_j \varphi - \chi\|_{L^1(dt)} \leq \|P_j - \chi/\varphi\|_{L^2(\gamma)} \longrightarrow 0.$$

$\square$

Proposition 11. *If (8) converges almost everywhere for each positive dilation of E and O , then*

$$\frac{1}{N} \int_0^N f(U_t y) g(V_t y) dt$$

converges almost everywhere for bounded measurable f, g .

Proof. Apply Lemma 8 to $H_N(y, t) = f(U_N t y) g(V_N t y)$. Its closed subspace contains all the Euler dilations by hypothesis, all polynomial Gaussian kernels by Lemma 9, and then χ by Lemma 10. Finally, the change of variables $s = Nt$ identifies the pairing with χ with the displayed time average. Only countably many kernels are used in each limiting argument, so no uncountable intersection of sets of full measure is required. $\square$

5 Suspension and exact reconstruction

Lemma 12. *On $Y = X \times [0, 1]^2$ with $\nu = \mu \otimes du \otimes dv$, the maps*

$$\begin{aligned} U_t(x, u, v) &= (T^{\lfloor u+t \rfloor} x, \{u+t\}, v), \\ V_t(x, u, v) &= (S^{\lfloor v+t \rfloor} x, u, \{v+t\}) \end{aligned}$$

define commuting, jointly measurable, measure-preserving $\mathbb{R}$ -actions. Here integer powers use the inverse transformations at negative times.

Proof. The floor identity

$$\lfloor u + s + t \rfloor = \lfloor u + s \rfloor + \lfloor \{u + s\} + t \rfloor$$

gives the group law for U , and the same argument gives it for V . The two maps act on different phase coordinates, and their actions on X commute because all integer powers of T and S commute. Joint measurability follows by partitioning the time-phase plane into the countably many measurable sets where its floor is a fixed integer.

For fixed t , write $t = q + a$ with $q \in \mathbb{Z}$ and $0 \leq a < 1$. Partition the u -interval into $[0, 1 - a)$ and $[1 - a, 1)$. On these two pieces U_t applies respectively T^q and T^{q+1} to x and translates u onto $[a, 1)$ and $[0, a)$. Invariance of μ and translation invariance of Lebesgue measure prove preservation of the product measure. The proof for V_t is identical. $\square$

Lift f, g to Y by ignoring the phase coordinates. Their interval-kernel average is

$$B_N(x, u, v) = \frac{1}{N} \int_0^N f(T^{\lfloor t+u \rfloor} x) g(S^{\lfloor t+v \rfloor} x) dt.$$

Lemma 13. *Let $w(u, v) = 3 - 6|u - v|$. For every bounded f, g , every x , and $N \geq 1$,*

$$\begin{aligned} \int_0^1 \int_0^1 w(u, v) B_N(x, u, v) dv du \\ = A_N(f, g)(x) + \frac{f(T^N x) g(S^N x) - f(x) g(x)}{2N}. \end{aligned} \tag{10}$$

Proof. For fixed n , abbreviate

$$\begin{aligned} a &= f(T^n x) g(S^n x), & b &= f(T^{n+1} x) g(S^n x), \\ c &= f(T^n x) g(S^{n+1} x), & d &= f(T^{n+1} x) g(S^{n+1} x). \end{aligned}$$

Splitting the time interval $[n, n+1]$ at $n+1-u$ and $n+1-v$ gives the cell integral

$$(1 - \max(u, v))a + (u - v)_+ b + (v - u)_+ c + \min(u, v)d.$$

The weighted square integrals of the four coefficients are

$$\begin{aligned} \int_{[0,1]^2} w(1 - \max(u, v)) &= \frac{1}{2}, & \int_{[0,1]^2} w(u - v)_+ &= 0, \\ \int_{[0,1]^2} w(v - u)_+ &= 0, & \int_{[0,1]^2} w \min(u, v) &= \frac{1}{2}. \end{aligned}$$

For example, on $u \geq v$ use $q = u - v$; the second integral is $\int_0^1 (3 - 6q)q(1 - q) dq = 0$. The first is $\int_0^1 (3 - 6q)(1 - q)^2 dq = 1/2$, after adding the two triangles. Symmetry proves the other two identities. Thus each cell contributes $(a + d)/2$. Summing in $n = 0, \dots, N - 1$ and dividing by N telescopes to (10). $\square$

Proposition 14. *The averages $A_N(f, g)$ converge almost everywhere whenever $f, g \in L^\infty(\mu)$.*

Proof. Use bounded measurable representatives. Apply Corollary 7 and Proposition 11 to the actions of Lemma 12. This proves convergence of B_N for ν -almost every (x, u, v) . By Fubini, for almost every x the convergence holds for almost every phase pair. Since

$$|B_N(x, u, v)| \leq \|f\|_\infty \|g\|_\infty, \quad |w(u, v)| \leq 3,$$

dominated convergence gives convergence of the left side of (10) for such x . The boundary term on its right side is bounded in modulus by $\|f\|_\infty \|g\|_\infty / N$ and tends to zero. Hence $A_N(f, g)(x)$ converges. The argument integrates over the whole phase square and does not restrict a product null set to a diagonal. $\square$

6 Extension to square-integrable functions

Proposition 15. *Fix two measure-preserving transformations T, S on a probability space. If $A_N(u, v)$ converges almost everywhere for every pair of bounded measurable functions u, v , then it converges almost everywhere for every $f, g \in L^2(\mu)$.*

Proof. Write $f_M = f \mathbf{1}_{\{|f| \leq M\}}$ and $g_M = g \mathbf{1}_{\{|g| \leq M\}}$, with M a positive integer, and set

$$P_N^T h(x) = \frac{1}{N} \sum_{n=0}^{N-1} h(T^n x).$$

Cauchy–Schwarz in the finite sum gives

$$\begin{aligned} |A_N(f, g) - A_N(f_M, g_M)| &\leq (P_N^T |f - f_M|^2)^{1/2} (P_N^S |g|^2)^{1/2} \\ &\quad + (P_N^T |f|^2)^{1/2} (P_N^S |g - g_M|^2)^{1/2}. \end{aligned} \quad (11)$$

Let $\mathcal{I}_T, \mathcal{I}_S$ be the invariant sigma-algebras and put

$$\begin{aligned} F_M &= \mathbb{E}(|f - f_M|^2 \mid \mathcal{I}_T), \quad G_M = \mathbb{E}(|g - g_M|^2 \mid \mathcal{I}_S), \\ F &= \mathbb{E}(|f|^2 \mid \mathcal{I}_T), \quad G = \mathbb{E}(|g|^2 \mid \mathcal{I}_S). \end{aligned}$$

The pointwise ergodic theorem identifies the four limits in (11) with F_M, G, F, G_M , respectively. These statements hold simultaneously outside a null set, since the family of truncation levels is countable. The functions F, G are finite almost everywhere.

The tail integrals tend to zero, so $\|F_M\|_1 + \|G_M\|_1 \rightarrow 0$. Choose a strictly increasing sequence M_j for which this sum is at most 2^{-j} . Tonelli's theorem gives $\sum_j (F_{M_j} + G_{M_j}) < \infty$ almost everywhere, and hence both F_{M_j} and G_{M_j} tend to zero there. By (11),

$$\limsup_{N \rightarrow \infty} |A_N(f, g) - A_N(f_{M_j}, g_{M_j})| \leq F_{M_j}^{1/2} G^{1/2} + F^{1/2} G_{M_j}^{1/2} \rightarrow 0.$$

Each truncated average converges by assumption. Its asymptotic oscillation is therefore zero. The triangle inequality bounds the asymptotic oscillation of $A_N(f, g)$ by twice the displayed error for every j . Letting $j \rightarrow \infty$ proves the Cauchy property and therefore convergence almost everywhere. $\square$

Proof of Theorem 2. Proposition 14, using the planar estimate proved in Proposition 1, gives convergence for bounded inputs. Proposition 15 extends it to $L^2 \times L^2$. On a probability space, L^∞ and L^3 are contained in L^2 , which proves all the stated conclusions. $\square$